

MPAS-URBAN QUICKSTART | GUIDE 3 OF 3

Run the HRAIN2025 case

Prepare the run directory, apply the 500 m mesh setting, and optionally use the HRAIN2025 NTDK cutoff modification.

OUTCOME You will launch the three-day HRAIN2025 atmosphere experiment and understand which settings are standard, case-specific, or required only for comparison with existing historical results.

AUTHOR LIU Zhuo

AFFILIATION The Hong Kong University of Science and Technology (HKUST)

Before you begin

Complete the recommended official MPAS tutorial and Guides 1–2. Confirm that the initial-condition file and `hk_hull_500m_graph.info.part.112` exist.

Choose the executable deliberately: standard `hkust-dev` for normal MPAS-Urban learning and new experiments, or the separate `ntdk-grid-cutoff` derivative for the HRAIN2025 cutoff configuration.

Step 1 | Prepare the run directory

FILES

```
export MPAS_SRC="$HOME/MPAS/HKUST-MPAS"
# For the optional HRAIN2025 cutoff build, use instead:
# export MPAS_SRC="$HOME/MPAS/HKUST-MPAS-NTDK"

export HRAIN_REPO="$HOME/HRAIN2025"
export CASE="$HOME/MPAS_CASES/HRAIN2025_30km_500m"

cd "$CASE"
ln -sf "$MPAS_SRC/atmosphere_model" .
cp "$HRAIN_REPO/config/namelist.atmosphere" .
cp "$MPAS_SRC/streams.atmosphere" .
cp "$MPAS_SRC"/*TBL . 2>/dev/null || true

ls -lh atmosphere_model *.init.nc  hk_hull_500m_graph.info.part.112
```

Also link or copy the physics lookup tables required by the selected source tree. Follow the official tutorial if your MPAS version stores these tables in a different directory.

Step 2 | Set the case time and mesh scale

REFERENCE NAMELIST

<https://github.com/Liuzh223/HRAIN2025/blob/main/config/namelist.atmosphere>

FILE TO EDIT `namelist.atmosphere`. Copy the repository version into the run directory and verify the following fields before launch.

CHANGE 1 - config_start_time: the first model time is 2025-08-03_00:00:00.

CHANGE 2 - config_run_duration: 3_00:00:00 covers the complete three-day case; the commented three-hour value is only a smoke test.

CHANGE 3 - config_dt: 6 seconds is the dynamics time step used by this experiment.

CHANGE 4 - config_len_disp: explicitly set 500.0 here in the running namelist. This is the first point in the tutorial where this case parameter is introduced.

NAMelist.ATMOSPHERE

```
&nhyd_model
  config_start_time = '2025-08-03_00:00:00'
  config_run_duration = '3_00:00:00'
  config_dt          = 6
  config_len_disp    = 500.0
/
```

REQUIRED CASE SETTING Keep config_len_disp = 500.0. It explicitly records the 500 m minimum mesh spacing used by this experiment and should not be omitted even when the mesh contains nominalMinDc.

Step 3 | Set the physics schemes and optional cutoff

FILE TO EDIT Continue editing the &physics group in namelist.atmosphere. The following suite is the explicit HRAIN2025 case configuration; verify every value rather than relying on source-tree defaults.

Physics suite used by HRAIN2025

The case uses WSM6 microphysics, New Tiedtke cumulus, Noah-MP land surface, YSU boundary layer and gravity-wave-drag options, cloud fraction, RRTMG longwave and shortwave radiation, the revised Monin-Obukhov surface layer, and MPAS-Urban. Use the same suite with either source tree; the standard hkust-dev build omits config_cu_disable_dx.

```
&physics
  config_microp_scheme = 'mp_wsm6'
  config_convection_scheme = 'cu_ntiedtke'
  config_lsm_scheme = 'sf_noahmp'
  config_pbl_scheme = 'bl_ysu'
  config_gwdo_scheme = 'bl_ysu_gwdo'
  config_radt_cld_scheme = 'cld_fraction'
  config_radt_lw_scheme = 'rrtmg_lw'
  config_radt_sw_scheme = 'rrtmg_sw'
  config_sfclayer_scheme = 'sf_monin_obukhov_rev'
  config_urban_physics = true
/
```

Optional HRAIN2025 cutoff derivative

Use the following only when atmosphere_model was compiled from Liuzh223/HKUST-MPAS-Official branch ntdk-grid-cutoff.

```
&physics
  config_convection_scheme = 'cu_ntiedtke'
  config_cu_disable_dx      = 10000.0
/
```

The value 10000.0 is a 10 km grid-spacing threshold. New Tiedtke convection is disabled where the local mesh spacing is below 10 km. The option default is 0.0, which leaves the cutoff disabled.

SOURCE NOTE ntdk-grid-cutoff is a case-author derivative, not the official HKUST-MPAS main branch. There is no separate shell command named cutoff.

Step 4 | Check the decomposition and streams

CHANGE 1 - namelist.atmosphere/&decomposition: use the prefix `hk_hull_500m_graph.info.part.` and launch 112 MPI tasks.

```
&decomposition
  config_block_decomp_file_prefix =
    'hk_hull_500m_graph.info.part.'
/
```

Use 112 MPI tasks so that the prefix resolves to `hk_hull_500m_graph.info.part.112`. In `streams.atmosphere`, point the input stream to the initial-condition file created in Guide 2 and select the desired output interval.

FILE TO EDIT `streams.atmosphere`. Set the input filename to `30km_500m.init.nc`, choose the history filename and output interval, and verify that all relative paths are resolved from the run directory.

Step 5 | Run and monitor

RUN

```
cd "$CASE"
export OMP_NUM_THREADS=1

mpiexec -n 112 ./atmosphere_model
```

CHECK

```
tail -n 60 log.atmosphere.0000.out
ls -lh history.*.nc 2>/dev/null || ls -lh *.nc
```

For an inexpensive first check, temporarily use a short `config_run_duration`, confirm that output is created, and then restore the intended three-day duration. Do not describe the short smoke test as the HRAIN2025 experiment result.

Final note | Comparing with existing HRAIN2025 results

The HRAIN2025 historical experiment results have not yet been publicly released. This section is needed only for a future direct comparison with those existing results. It is not required for learning MPAS or for a new independent simulation.

The reference result source is located in Liuzh223/HKUST-MPAS, branch rainfall. For an immutable reference, record the exact commit used for the comparison.

SOURCE LOCATION

```
src/core_atmosphere/physics/physics_wrf/module_sf_urban.F
```

Inside subroutine SFCDIF_URB, inspect both ZT updates: the first calculation after USTAR is obtained and the update inside the stability iteration. For consistency with the existing HRAIN2025 result calculation, the final active expression at both locations is:

$$ZT = 0.1 * Z0$$

After changing the source, rebuild atmosphere_model before running the comparison. Do not confuse the target expression with Z0HC_TBL = 0.1 * Z0C_TBL; that is a different calculation.

For a new independent experiment, keep the ZT implementation supplied by the selected HKUST-MPAS version and record the repository, branch, and commit. Do not modify ZT merely to complete the tutorial.

Completion checklist

- config_start_time is 2025-08-03_00:00:00.
- config_run_duration is 3_00:00:00 for the full experiment.
- config_len_disp is explicitly 500.0.
- config_cu_disable_dx is present only for a verified cutoff-capable build and is 10000.0 for this case.
- The MPI task count matches .part.112.
- ZT is changed only when directly comparing with the existing HRAIN2025 historical results.